Review Notes

Week 8

1 LTI Systems

LTI Systems properties

Linearity

A system T is **linear** if for every two inputs $x_1[n], x_2[n]$ and scalars a, b,

$$T\{ax_1[n] + bx_2[n]\} = aT\{x_1[n]\} + bT\{x_2[n]\}. \tag{1}$$

This is the superposition principle.

Example (linear but not time-invariant):

$$T\{x[n]\} = nx[n].$$

Time invariance

A system T is **time-invariant** if a shift in the input produces the same shift in the output. If $T\{x[n]\} = y[n]$, then for every integer n_0 :

$$T\{x[n - n_0]\} = y[n - n_0]. \tag{2}$$

Example (time-invariant but not linear):

$$T\{x[n]\} = x[n] + 1.$$

A system is LTI only if it is both linear and time-invariant.

Why are LTI systems so useful?

For an LTI system, once we know the response to $\delta[n]$, we know the response to *every* input.

Let $h[n] = T\{\delta[n]\}$ be the impulse response. Any signal can be written as:

$$x[n] = \sum_{k=-\infty}^{\infty} x[k]\delta[n-k]. \tag{3}$$

Using linearity and time invariance:

$$\begin{aligned} T\{x[n]\} &= T\{\sum_{k=-\infty}^{\infty} x[k]\delta[n-k]\} \\ &= \sum_{k=-\infty}^{\infty} x[k]T\{\delta[n-k]\} \\ &= \sum_{k=-\infty}^{\infty} x[k]h[n-k] \end{aligned} \tag{4}$$

Therefore,

$$y[n] = x[n] * h[n]. \tag{5}$$

And this holds only for LTI systems.

Example :

Suppose an LTI system has impulse response

$$h[n] = \left(\frac{1}{2}\right)^n u[n] \tag{6}$$

and the input is

$$x[n] = \delta[n] + 2\delta[n-1]. \tag{7}$$

Then, by linearity and time invariance,

$$\begin{aligned} y[n] &= T\{\delta[n] + 2\delta[n-1]\} \\ &= T\{\delta[n]\} + 2T\{\delta[n-1]\} \\ &= h[n] + 2h[n-1] \end{aligned} \tag{8}$$

Hence,

$$y[n] = \left(\frac{1}{2}\right)^n u[n] + 2\left(\frac{1}{2}\right)^{n-1} u[n-1]. \tag{9}$$

2 Exercises

Exercise 1

A discrete-time system is defined by an input-output relationship $y[n] = x[-n]$. Which of the following claims about the system is true?

(a) The system is linear

(b) The system is time-invariant

(c) The system is causal

(d) The system is LTI

Solution: The answer is (a). The system is linear, but not time-invariant or causal. Since it is not time-invariant, it is also not LTI.

Exercise 2

An LTI system is given by the difference equation:

$$y[n] = \lambda_1 y[n-1] + \lambda_2 x[n], \tag{10}$$

where $\lambda_1, \lambda_2 > 0$.

Assume that the system satisfies the condition of initial rest. That is, if $x[n] = 0$ for all $n < n_0$, then $y[n] = 0$ for all $n < n_0$. Find the impulse response of the system. Is this system an FIR or an IIR filter?

Solution: The impulse response of the system can be found by induction. First, we use the initial rest condition. Since $\delta[n] = 0$ for all $n < 0$, $h[n] = 0$ for all $n < 0$. Then,

$$\begin{aligned} h[0] &= \lambda_1 h[-1] + \lambda_2 \delta[0] = \lambda_2 \\ h[1] &= \lambda_1 h[0] + \lambda_2 \delta[1] = \lambda_1 \lambda_2 \end{aligned} \tag{11}$$

Now assume:

$$h[n] = \lambda_1^n \lambda_2 u[n]. \tag{12}$$

Clearly, the assumption holds for all $n \leq 1$. Then, for all $n \geq 1$, by induction,

$$\begin{aligned} h[n+1] &= \lambda_1 h[n] + \lambda_2 \delta[n+1] \\ &= \lambda_1 (\lambda_1^n \lambda_2 u[n]) \\ &= \lambda_1^{n+1} \lambda_2. \end{aligned} \tag{13}$$

Thus, the impulse response is

$$h[n] = \lambda_2 \lambda_1^n u[n]. \tag{14}$$

The system is an IIR filter.

Alternate solution: Computing the Z-transform of both sides of equation (10), we get

$$Y(z) = \lambda_1 z^{-1} Y(z) + \lambda_2 X(z).$$

Rearranging in order to get $H(z) = \frac{Y(z)}{X(z)}$, we have

$$Y(z) - \lambda_1 z^{-1} Y(z) = \lambda_2 X(z),$$

and thus

$$Y(z) = \frac{\lambda_2}{1 - \lambda_1 z^{-1}} X(z).$$

This yields

$$H(z) = \frac{\lambda_2}{1 - \lambda_1 z^{-1}} \implies h[n] = \lambda_2 \lambda_1^n u[n]$$

by causality.